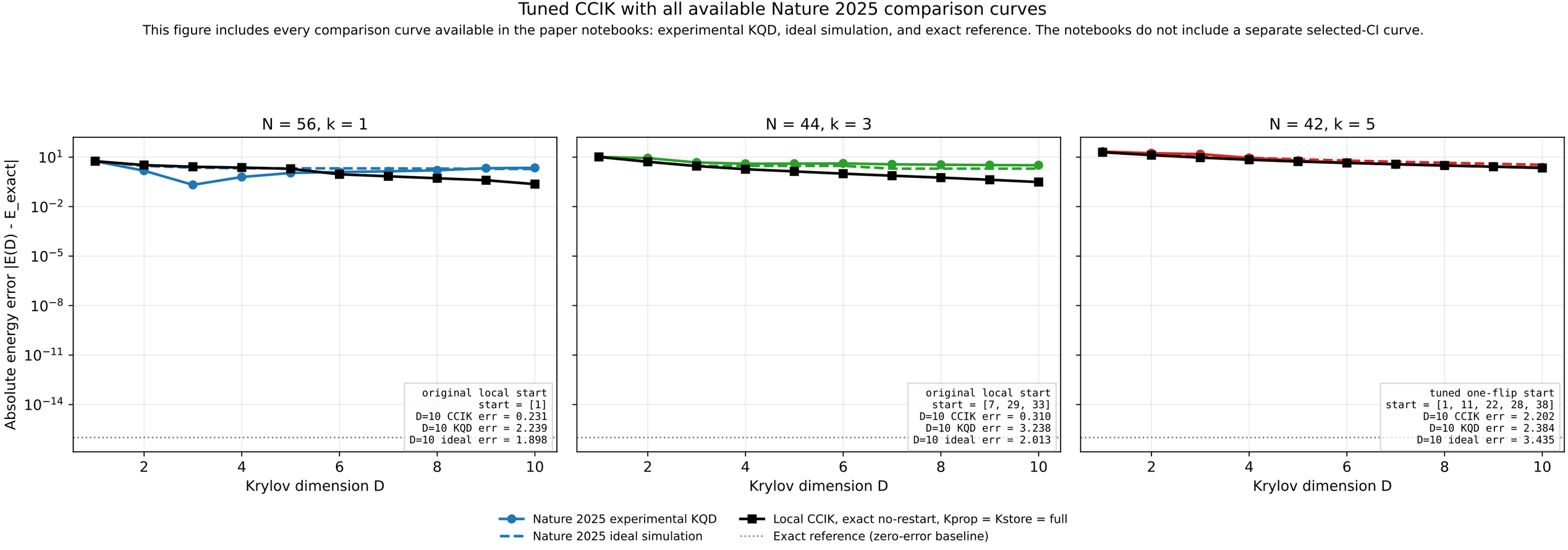
Tuned CCIK with all available Nature 2025 comparison curves
This figure includes every comparison curve available in the paper notebooks: experimental KQD, ideal simulation, and exact reference. The notebooks do not include a separate selected-CI curve.
N = 56, k = 1
N = 44, k = 3
N = 42, k = 5
Absolute energy error |E(D) - E_exact|
Krylov dimension D
original local start
start = [1]
D=10 CCIK err = 0.231
D=10 KQD err = 2.239
D=10 ideal err = 1.898
original local start
start = [7, 29, 33]
D=10 CCIK err = 0.310
D=10 KQD err = 3.238
D=10 ideal err = 2.013
tuned one-flip start
start = [1, 11, 22, 28, 38]
D=10 CCIK err = 2.202
D=10 KQD err = 2.384
D=10 ideal err = 3.435
Nature 2025 experimental KQD
Nature 2025 ideal simulation
Local CCIK, exact no-restart, Kprop = Kstore = full
Exact reference (zero-error baseline)